

Auditor Replay for Anonymous-DHT Receipt Line Items

A Mechanical Checklist for Digest-Bound Claims (Series Synthesis)

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Abstract

This archive publishes privacy claims as compact receipts: an envelope plus line items with digest-bound replay hooks. Many papers implicitly assume an auditor can “replay the accountant” from a receipt and an artifact bundle, but they do not spell out the mechanical steps. This note is the canonical replay checklist. It standardizes: (i) what objects an auditor needs, (ii) what must be checked before doing any privacy math, and (iii) what a successful replay verdict should record. It is intentionally not a privacy proof; it is a verification workflow contract so other papers can stay short.

Scope. This note standardizes *mechanical replay*: digest checks, envelope matching, log anchoring, and plan-spec replay. It does *not* define endpoint metrics, witness tiers, or composition theorems; those are imported.

Imports. Receipt schema (envelope + line items) is owned by Synthesis 12.[1] Plan/state drift and digest binding rules are owned by Synthesis 18.[2] Horizon- n accounting rules (retries/state) are owned by Synthesis 19.[3] Conditional per-step certificate shape for Regime II is owned by Synthesis 21.[6] ABOM/OINL packaging is owned by Anonymity C.[4] Transparency primitives (append-only logs, inclusion/consistency) are owned by Synthesis 9.[5] A concrete end-to-end instantiation appears in the worked example (Synthesis 17).[7]

Exports. A checklist (section 2) and a small *replay verdict* record format (section 3) that auditors and tooling can emit.

1 Objects an auditor needs (minimal set)

An auditor should be able to do a replay using only public objects plus an on-request artifact bundle. The minimal set is:

- **Receipt** R (envelope + line items).[1]
- **ABOM** A (manifest committing to claim semantics and artifact digests).[4]
- **UVI/log anchor** U (receipt digest + inclusion/consistency proofs), or an equivalent transparency checkpoint chain.[5]
- **Registries** used to bind human labels to digest ids (ENF/TW/State registries, when the receipt uses digest ids).

- **On-request artifact bundle** referenced by each line item’s `artifact_bundle` (the omitted proofs, logs, scripts, certificates).

Remark 1 (Why ABOM is an input, not optional). *In this archive, the receipt is intentionally small. Without ABOM, an auditor cannot tell whether the artifact bundle presented on request is the bundle the publisher committed to when publishing the receipt. ABOM provides the digest commitments that make “send me the artifacts” checkable.*[4]

2 Mechanical replay checklist

The checklist below is intentionally ordered: fail fast on anchoring and digest mismatch before doing any privacy-domain computation.

2.1 Step 0: log anchoring and signature hygiene

1. **Compute receipt digest.** Compute $h(R)$ under the canonical digest rule the series uses.
2. **Verify inclusion.** Verify the Merkle inclusion proof that commits $h(R)$ under the stated checkpoint.[5]
3. **Verify consistency context.** Verify the checkpoint chain or consistency proof from the auditor’s last checkpoint.[5]
4. **Verify signatures.** Verify the receipt/ABOM signatures under the declared keys (release key, notarization key, or both), as specified by the publication layer.[4]

2.2 Step 1: envelope matching (drift guard)

Receipts are compared across releases by their envelope. Before replaying any line item, check that the published objects agree on the envelope IDs:

1. **Receipt envelope.** Read `claim_id`, `tw_id`, and `state_decl_id` from the receipt envelope.[1]
2. **ABOM claim fields.** Confirm ABOM/OINL pins the same claim family and threat-window semantics.
3. **UVI pointer.** Confirm the user-facing pointer (if present) names the same receipt digest and (at least) the same `claim_id` and `exposure_nf_id`. [5]

If any envelope ID differs, the auditor should treat the objects as distinct claims and avoid “diffing” budgets as if they were comparable.[2]

2.3 Step 2: line-item integrity and replay-hook checks

For each line item ℓ in R :

1. **Projection binding.** Confirm `exposure_nf_id` is a digest id (or label resolvable to one) and matches the extractor/schema the mechanism paper claims to budget.[1]
2. **Threat-window binding.** Confirm `tw_id` matches the receipt envelope or is explicitly overridden (rare; should be justified).[1]

Field	Meaning
<code>receipt_digest</code>	Digest of the exact receipt object audited
<code>envelope</code>	<code>claim_id</code> , <code>tw_id</code> , <code>state_decl_id</code>
<code>line_item_verdicts</code>	For each line item: <code>exposure_nf_id</code> , <code>plan_id</code> , <code>plan_spec_id</code> , recomputed scalar(s), and pass/fail
<code>artifact_bundle_digests</code>	Digests of the artifact bundles actually replayed
<code>log_checkpoint</code>	The checkpoint used for inclusion/consistency
<code>notes</code>	Any regime/assumption notes (e.g., independence vs stateful accounting)

Table 1: A minimal replay verdict is itself a small diffable object.

3. **Replay hook is digest-bound.** Confirm `replay_hook` includes `plan_id` and `plan_spec_id`, and that `plan_spec_id=sha256(canon(plan_spec))` for the on-request plan spec.[2]
4. **State surface scope.** If the line item depends on state (explicitly or by regime), confirm the relevant `state_decl_id` is present (either at the envelope or inside the line item) and matches the declared state scope.[3]

2.4 Step 3: artifact-bundle commitment check

For each referenced `artifact_bundle`:

1. **Resolve bundle.** Use the public bundle map (if provided) to list required artifacts for replay.
2. **Verify digests.** Confirm each delivered artifact matches the ABOM-committed digest list.
3. **Verify provenance.** If the plan spec requires notarization or transparency anchoring (e.g., a plan log inclusion proof), verify those proofs before using the artifact as evidence.[5]

2.5 Step 4: replay the accountant

Finally, run the replay plan described by `plan_spec`:

1. **Recompute the published quantity.** Using the plan spec, recompute the line item’s target scalar (e.g., per-lookup MaxL bound after attenuation, selection-tax term, or a TV dual-certificate bound).
2. **Check interpretation.** Confirm the receipt’s `budget_type` matches the endpoint metric meaning (imported from the endpoint bridge) and that the unit is consistent with `usage` (per lookup vs horizon).[3]
3. **Record regime.** If a horizon claim is made (e.g., “ $Q\times$ ” or a published total), record which regime of Synthesis 19 was used and which independence/state assumptions were relied on.[3]

3 Replay verdict record (what to output)

To keep audits diffable across time, an auditor should emit a small verdict record. A minimal record is:

Remark 2 (Why the verdict is part of the publication story). *A replay verdict is a way to publish that the replay was performed without publishing the full private artifact archive. It is compatible with the source-only posture: the verdict can be public and small; the artifacts remain available on request.*

4 Worked example pointer

The worked-example bundle (Synthesis 17) provides concrete instances of every checklist step: envelope IDs, registries, replay plans, support-bundle maps, and a deterministic validation report.[7] The purpose of this note is to make that workflow citable without requiring every paper to point directly at the worked example.

References

- [1] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Series synthesis note (Synthesis 12), 2026. (This archive.)
- [2] Anonymous. Plan and State Equivalence: Digest-Bound Replay Hooks and Drift Taxonomy. Series synthesis note (Synthesis 18), 2026. (This archive.)
- [3] Anonymous. Receipt Accounting and Horizon Composition for Anonymous-DHT Budgets. Series synthesis note (Synthesis 19), 2026. (This archive.)
- [4] Anonymous. ABOM/OINL: Anonymity Bills of Materials and Odds-Inflation Nutrition Labels. Anonymity series Paper C, 2026. (This archive.)
- [5] Anonymous. Transparency Primitives for Receipts and Certificates. Series synthesis note (Synthesis 9), 2026. (This archive.)
- [6] Anonymous. Conditional Per-Step Budget Certificates: A Receipt-Grade Interface for Adaptive Horizon Claims. Synthesis 21, The-One-True-Archive (2026).
- [7] Anonymous. Worked Example: Receipt Interlock for an Anonymous-DHT Reader-Privacy Claim. Series synthesis note (Synthesis 17), 2026. (This archive.)